

## ALGORITHM: SPY Volatility Estimation Using Moving Average and EWMA

### INPUT:

- \* SPY daily adjusted closing prices
- \* Start date = 08/01/2021
- \* End date = 07/30/2026
- \* Look-back period  $n = 100$
- \* EWMA decay factor  $\lambda = 0.94$
- \*  $\Delta = 1/252$

### OUTPUT:

- \* Annualized volatility from 100-day Moving Average method
- \* Annualized volatility from EWMA method
- \* Plot of both volatility estimates on the same figure

### BEGIN

1. Download daily SPY adjusted closing prices.
2. FOR each trading day  $t$  from the second observation to the last observation:  
    Compute daily log return:  
     $r_t = \ln(P_t / P_{(t-1)})$   
    END FOR
3. Remove the first missing return observation.
4. Initialize Moving Average volatility series.
5. FOR each trading day  $t$  starting from day  $n$ :  
    Select the most recent  $n = 100$  log returns.  
    Compute sample variance:  
     $\text{Var\_MA}(t) = \text{variance of previous 100 returns}$   
  
    Compute annualized volatility:  
     $\text{Vol\_MA}(t) = \sqrt{\text{Var\_MA}(t) / \Delta}$   
    END FOR
6. Compute the sample variance of the first 100 log returns.
7. Set:  
     $\text{Var\_EWMA}(100) = \text{initial sample variance}$
8. FOR each trading day  $t$  starting from day 101:  
    Update EWMA variance:  
     $\text{Var\_EWMA}(t) =$   
     $\lambda \times \text{Var\_EWMA}(t-1)$   
     $+ (1 - \lambda) \times r_{(t-1)}^2$   
  
    Compute annualized volatility:

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Vol_EWMA(t) = sqrt(Var_EWMA(t) / Delta)
END FOR
```

9. Create a single figure.
10. Plot Vol\_MA against date.
11. Plot Vol\_EWMA against date on the same figure.
12. Add:

- \* Title
- \* X-axis label: Date
- \* Y-axis label: Annualized Volatility
- \* Legend for Moving Average and EWMA

13. Display the figure.

### **ALGORITHM: Fit SPY Daily Returns with GARCH(1,1)**

#### INPUT:

- SPY daily closing prices
- Start date = 06/01/2025
- End date = 05/31/2026

#### OUTPUT:

- Estimated GARCH(1,1) parameters
- Annualized conditional volatility series

#### BEGIN

1. Download SPY daily closing prices over the specified period.
2. FOR each trading day t from the second observation to the last observation:  
    Compute daily log return:

$$r_t = \ln(P_t / P_{(t-1)})$$

END FOR

3. Remove the first missing return observation.
4. Convert log returns to percentage returns:

$$R_t = 100 \times r_t$$

5. Specify a GARCH(1,1) model:

$$\begin{aligned} \text{Variance}_t = & \\ & \text{omega} \\ & + \alpha \times \text{Error}_{(t-1)}^2 \\ & + \beta \times \text{Variance}_{(t-1)} \end{aligned}$$

6. Assume:

- Constant mean return
- Normally distributed errors

7. Estimate the model parameters using the daily percentage returns.

8. Obtain the estimated parameters:

mu  
omega  
alpha  
beta

9. Extract the fitted daily conditional volatility from the GARCH model.

10. Convert conditional volatility from percentage units back to decimal units:

Daily Volatility =  
Conditional Volatility / 100

11. Set:

Delta = 1 / 252

12. Annualize the daily conditional volatility:

Annualized Volatility =  
Daily Volatility / sqrt(Delta)

13. Plot annualized conditional volatility against date.

14. Add:

- Plot title
- Date label on x-axis
- Annualized Volatility label on y-axis
- Legend

15. Display the estimated GARCH(1,1) model results.

### **ALGORITHM: Simulate GARCH(1,1) Volatility Paths and Compare with MA and EWMA**

INPUT:

- Previously fitted GARCH(1,1) model
- Previously calculated 100-day Moving Average volatility
- Previously calculated EWMA volatility
- Forecast period = 06/01/2026 to 07/30/2026
- Number of simulation paths M = 100

OUTPUT:

- 100 simulated GARCH volatility paths

- Mean simulated GARCH volatility path
- Analytical GARCH volatility forecast
- Comparison plot with MA and EWMA volatility

BEGIN

1. Select MA volatility estimates over the forecast period.
2. Select EWMA volatility estimates over the forecast period.
3. Use the dates from the forecast period as the future trading dates.
4. Set:  
 $H$  = number of trading days in the forecast period  
 $M = 100$
5. Using the previously fitted GARCH(1,1) model,  
generate  $M$  simulated variance paths over  $H$  future days.
6. FOR each simulated path  $m = 1$  TO  $M$ :

FOR each forecast day  $h = 1$  TO  $H$ :

Obtain the simulated conditional variance  
from the fitted GARCH model.

Convert variance to volatility:

$$\text{Volatility}_{(h,m)} = \sqrt{\text{Variance}_{(h,m)}}$$

END FOR

END FOR

7. Convert simulated GARCH volatility from percentage units  
to decimal units.
8. Annualize each simulated volatility:

$$\text{Annualized Volatility}_{(h,m)} = (\text{Volatility}_{(h,m)} / 100) \times \sqrt{252}$$

9. FOR each forecast day  $h$ :

Compute the average volatility across all  $M$  paths:

$$\begin{aligned} \text{Mean\_GARCH}(h) = & \text{average of Annualized Volatility}_{(h,m)} \\ & \text{for } m = 1, \dots, M \end{aligned}$$

END FOR

10. Using the same fitted GARCH model,  
calculate the analytical multi-step variance forecast  
over H days.

11. Convert the analytical variance forecast to  
annualized volatility:

$$\text{Analytical\_GARCH}(h) = (\sqrt{\text{Forecast Variance}_h} / 100) \times \sqrt{252}$$

12. Create one figure.

13. FOR each simulated GARCH path m = 1 TO M:

Plot the annualized GARCH volatility path  
using a thin transparent line.

END FOR

14. Plot the mean of the 100 simulated GARCH paths.

15. Plot the analytical GARCH volatility forecast.

16. Plot the 100-day Moving Average volatility  
over the same forecast period.

17. Plot the EWMA volatility  
over the same forecast period.

18. Add:

- Title
- Date label on x-axis
- Annualized Volatility label on y-axis
- Legend

19. Display the figure.